CLASSIFICATION

The Petroleum industry generally classifies crude oil by the geographic location it is produced in, E.g West Texas Intermediate, Brent, or Oman. API gravity (an oil industry measure of density) and its Sulphur content.

A crude oil ~~oit~~ with low density is considered "light" and "heavy" if it has high density and may be considered "sweet" if it contains little sulphur or "sour" if it contains much sulphur.

The geographic location is important because it affects transportation costs to the refineries. Light crude oil is more desirable than heavy oil since it produces a higher yield of gasoline, while sweet oil commands a higher price than sour because it has fewer environmental problems and requires less refining to meet sulphur standards imposed ~~on~~ on fuels in consuming countries.

Crude oil from an area in which the oil's molecular characteristics have been determined and the oil has been classified are used as pricing references throughout the world. Some of the common reference crudes are:

* West Texas Intermediate (WTI); A very high-quality, sweet, light oil delivered at Oklahoma for North America.

* Brent blend: This comprises 15 oils from various fields of the North Sea. The oil is landed at Sullom Voe Terminal in Shetland. Oil production from Europe, Africa and ~~Middle East~~ middle eastern oil flowing west has this as their benchmark.

* Dubai-Oman: This is used as benchmark for Middle East sour crude oil flowing to the Asia-Pacific region.

* Tapis: From Malaysia, used as a reference for light far east oil.

* Minas: From Indonesia, used as a reference for heavy far east oil.

* OPEC Reference Basket: A weighted average of oil blends from various OPEC Countries.

* Midway Sunset Heavy: by which heavy oil in California is priced.

* Western Canadian Select: The benchmark crude oil for emerging heavy, high ~~acidic~~ crudes.

THEORIES OF THE ORIGIN OF PETROLEUM

There are many theories about the origin of petroleum. But the most widely accepted theory is the biological theory which is also called the modern theory. The prominent among these theories are: Carbide theory, Organic or Engler's theory and the modern theory.

1. Carbide theory: According to this theory the origin of petroleum is ascribed to pure inorganic sources. It was first put forward by Mendeleeff and later supported by Moissan, Sabatier and Senderens. According to this theory the molten metals in the hot interior of the earth underwent reaction with local deposits under high temperature and pressure to yield carbides. Then, these carbides reacted with steam or water under high temperature and pressure to yield mixture of hydrocarbons — saturated and unsaturated.

1st stage: $Ca + 2C \longrightarrow CaC_2$ (Calcium carbide)

$4Al + 3C \longrightarrow Al_4C_3$ (aluminium carbide)

$Mg + 2C \longrightarrow MgC_2$ (magnesium carbide)

2nd stage: $CaC_2 + H_2O \longrightarrow Ca(OH)_2 + C_2H_2$ (Acetylene)

$Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4$ (methane)

3rd stage: In the presence of metal catalyst and under conditions high temperature and pressure, the unsaturated hydrocarbons, undervent hydrogenation, polymerization, isomerization and alkalation to yield a variety of saturated or unsatud hydroc...

Example: i, $C_2H_2 + H_2 \rightarrow C_2H_4$ (Ethylene)

ii, $C_2H_4 + H_2 \rightarrow C_2H_6$ (Ethane)

iii, $CH_3CH=CH-CH_3 + H_2 \rightarrow CH_3-CH_2-CH_2-CH_3$ (Butane

iv, $C_2H_4 \rightarrow CH_3-CH=CH-CH_3$ (2-Butene)

v, $3C_2H_4 \rightarrow C_6H_{12}$ (Cyclohexane)

vi, $3C_2H_2 \rightarrow C_6H_6$ (Benzene)

2. Organic theory or Engler's theory: Engler (1900)
Postulated that petroleum is formed by the decay
and decomposition of animals, preferably of Sea
origin, under high temperature and pressure.
He argued that if volcanic eruption occurs by the Sea
side then poisonous gases like sulphurdioxide(SO_2)
given out by the volcano may dissolve in sea water
and fishes and other marine animals in the sea
would therefore, die because of suffocation. In due
course of time this may cause huge accumulation of
dead marine animals and if somehow by (earth-
quakes) this zone becomes covered by the earth's hot crust
then because of high pressure and temperature inside,
decomposition may start in the formation of petroleum.
He supported his theory by carrying out destructive
decomposition of fish oil and other animal fats
under high pressure and temperature which produ-
ced a product similar to natural petroleum con-
taining compounds of N and S and some optically
active compounds as well.
This theory was supported by the following facts:
1, Presence of brine or sea water together with petroleum.
2, Presence of ~~active~~ optically active comp'ds and those
of N and S.
3, Presence of fossils in the petroleum area.

However it fails to explain the following facts:

1, Presence of Chlorophyll c

2, Presence of Coal deposits found near the oil fields.

3, High resin contents of certain oils.

3. Modern Theory: According to this theory petroleum
is believed to be of biological origin formed by
the decay and decomposition of marine animals
as well as that of vegetables of the pre-historic forests.
These marine animals and plants got embedded or trapp-
ped inside the earth millions of years ago due to
erosions, upheavals or earthquakes and are subjected
to the effect of moisture, high temperature and pressure
and a little air and thus changed into carbon-
aceous residues under the influence of radioactive
substances or by bacterial agency. Thus, it is possible
that petroleum was derived from marine organisms
while coal was from plants on earth.

The inorganic substances like (S, H, N, O, etc)
present in the living organisms got converted either
into inorganic gases or into mineral matter while
most of the carbon got converted into natural gas, petroleum
and solids such as (Shale, Diamond, Coal, etc).

The modern theory satisfactorily explains the afore
mentioned facts ~~etc.~~ in addition to the following:
i, It explains the presence of brine and coal near the petr-
oleum. Brine was on the basis of animal ~~origin~~ and vege-
table origin.
ii, The presence of optically active compounds, N and S – compounds,
chlorophyll and haemin compounds makes the oil of
organic origin.

iii, Presence of resins reveals that petroleum must have been formed from plant substances.

iv, Distillation of fish blubber yielded a product which resembles petroleum.

COMPOSITION OF PETROLEUM

The composition of petroleum varies from place to place and also from well to well. It's colour varies from yellow to black through reds and generally consists of alkanes from C_1 to C_{40}, Cyclopentane and cyclohexane derivatives called naphthenes in petroleum technology and aromatic hydrocarbons - which are usually present in small quantities. However, ~~some~~ petroleum obtained from some other sources consists of appreciable amounts of aromatic hydrocarbons and such a petroleum forms the principal source of supply of such comp'ds. Besides these comp'ds it also contains small quantities of nitrogen, oxygen or sulphur containing comp'ds.

Since crude oils or petroleum are comp'ds that vary in appearance and composition. Average crude oil composition by weight is 84% C, 14% H, 1-3% S and less than 1% each of nitrogen, oxygen, metals and salts.

They are distinguished as sweet or sour depending upon the sulphur content, which may be in the form of hydrogen sulphides. Crude oils with high sulphur content are called sour and those with less sulphur are called sweet.

Sweet crude is highly sort after because of the less sulphur in it. The operational costs for processesing it is very less and so also its environmental acceptability which is very high. Other classifications of crude oils could be based on Industrial need, origin, Economic, Chemical and Technological considerations.

Four different types of hydrocarbon molecules appear in crude oil. The relative % of each varies from oil to oil determining the ppties of each oil.

Hydrocarbon	Average %	Range
Alkanes (Paraffins)	30%	15-60%
Naphthenes (cycloalkanes) (Naphthenics)	49%	30-60%
Aromatics	15%	3-30%
Asphaltics	6%	Remainder, i.e; 10%

Petroleum being a mixture of a very large number of different hydrocarbons, the most commonly found molecules are alkanes - paraffins, cycloalkanes (naphthenes), aromatics and the more complicated chemicals like asphaltenes. Each petroleum variety has a unique mix of molecules which define its physical and chemical ppties, like colour and viscosity.

The alkanes are saturated hcs with straight or branched chains which contain only C and H and have the general formula C_nH_{2n+2}. They generally have from 5-40 carbon atoms per molecule.

The alkanes from pentane C_5H_{12} to octane C_8H_{18} are refined into gasoline, the ones from nonane C_9H_{20} to hexadecane $C_{16}H_{34}$ into diesel fuel; kerosene & jet fuel.

Alkanes with more than 16-C atoms can be refined into
fuel oil and lubricating oil. At the heavier end of th
range, Paraffin wax is an alkane with approx. 25-C
atoms, while asphalt has up to 35 and above, althou
these are usually cracked by modern refineries into
more valuable prdts.

The shortest molecules, those with 4 or fewer carbon ato
are in gaseous state at room temp. They are the Petr
oleum gases. Depending on demand and cost of recove
these are either flared off, sold as liquefied petroleum
gas (LPG) under pressure, or used to power the refi-
nery's own burners.

Butane (C_4H_{10}), can be blended into the gasoline pool at
high rates because its high vapour pressure assists with
Cold starts. Liquefied under pressure slightly above
atmospheric, it is best known for powering cigarette
lighters but it is also a main source of fuel for
Many developing countries.

Propane can be liquefied under modest pressure
and is consumed by applications relying on
Petroleum for energy from cooking to heating to
transportation.

The cycloalkanes, also known as the naphthenes,
are saturated hydrocarbons which have one or
more carbon rings to which hydrogen atoms are
attached according to the formula C_nH_{2n}.
They have similar properties to alkanes but
have higher boiling points.

The aromatic hydrocarbons are unsaturated HCs which have one or more ~~carbon~~ planar six-carbon rings called benzene rings, to which hydrogen atoms are attached with the formula C_nH_n.
They tend to burn with a sooty flame and many have a sweet aroma.
These different molecules are separated by fractional distillation at an oil refinery to produce petrol, jet fuel, kerosene and other hydrocarbons.
For e.g 2,2,4-trimethylpentane (Isooctane), widely used as gasoline reacts with oxygen exothermically

$$2C_8H_{18(l)} + 25O_{2(g)} \rightarrow 16CO_{2(g)} + 18H_2O_{(g)}$$

$$\Delta H = -5.51\ MJ/Mol$$

ANALYSIS OF OIL or Crude oil Assay

The number of various molecules in an oil sample can be determined in the laboratory.
First, the molecules are typically extracted in a solvent, then separated in a gas chromatograph and finally determined with a suitable detector or a mass spectrometer.
A crude oil assay is essentially the xcal evaluation of crude oil feedstocks by petroleum refining laboratories or testing labs.
Each oil type has unique molecular, chemical characteristics. No crude oil type is identical and there are important differences in crude oil quality.

~~INTRODUCTION~~

The result of crude oil testing assay provide refiners extensive detailed hydrocarbon analysis data and to oil traders and producers alike.

Assay data help refineries ~~to~~ determine if a crude oil is compatible for a particular petroleum refinery or if the crude oil could cause yield, quality, production, environmental and other problems. It can be an inspection assay or comprehensive assay. Testing can include crude oil characterization and the various boiling range fractions produced from physical or simulated distillation by various ~~producess~~ procedures.

Information obtained from the petroleum assay is used for detailed refinery engineering and client marketing purposes; hence, feed stock assay data are an important tool in the refining processes.